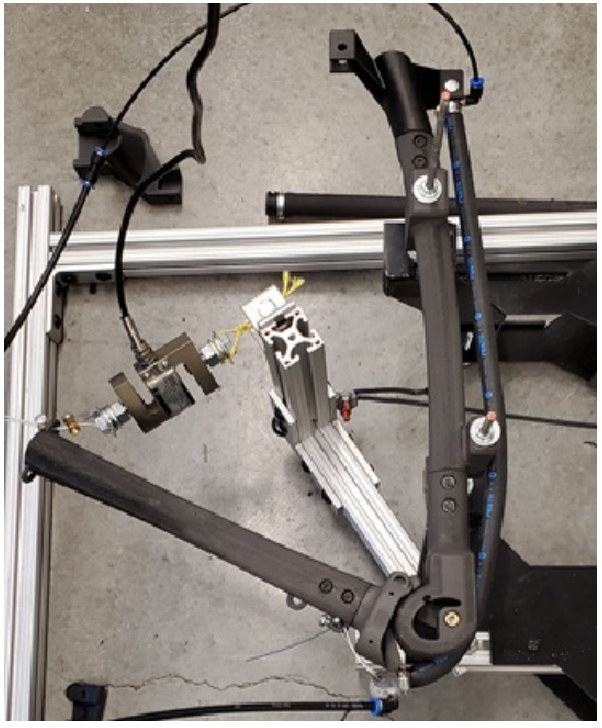
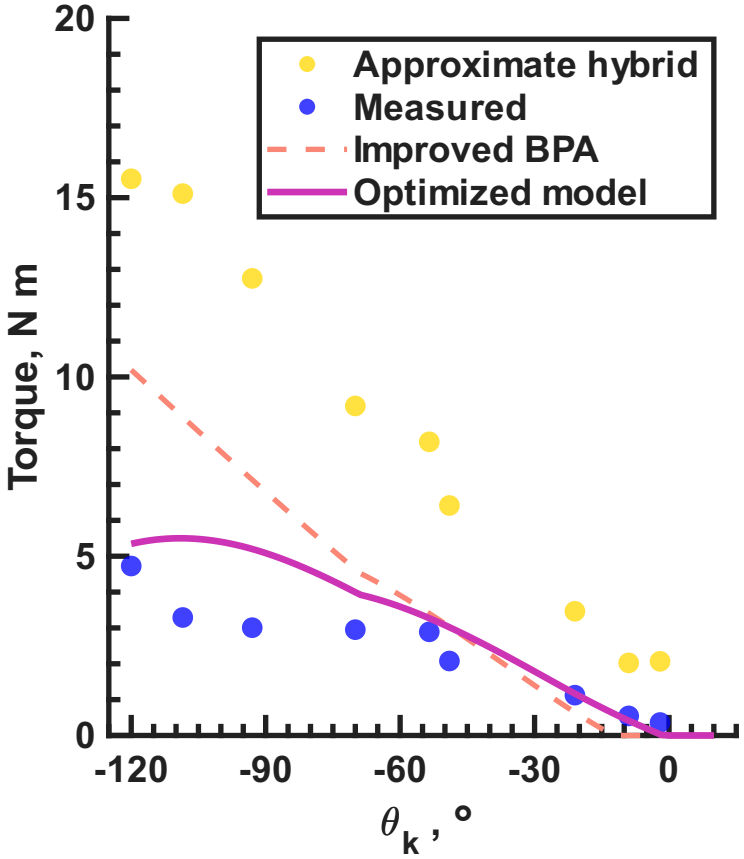
(A)



(B)


Approximate hybrid
Measured
Improved BPA
Optimized model
Torque, N m
20
15
10
5
0
-120
-90
-60
-30
0
θk , °